Pr. 9.

(8.33)

(1) $$x\cdot(\pi - x) = \underbrace{\frac{4}{\pi}\cdot\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^3}\sin(nx)}_{RHS}.$$

(2) $$RHS = \frac{4}{\pi}\cdot\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^3}\sin(nx)$$

(3) $\boxed{n = 2k}$ $k = 1, 2..$

(4) $$RHS = \frac{4}{\pi}\cdot\sum_{k=1}\frac{1-(-1)^{2k}}{(2k)^3}\sin(nx) = \frac{4}{\pi}\sum 0 = 0.$$

(5) $\boxed{n = 2k-1}$ $k = 1, 2..$

(6) $$RHS = \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{1-(-1)^{2k-1}}{(2k-1)^3}\sin[(2k-1)x] = \frac{8}{\pi}\cdot\sum_{k=1}^{\infty}\frac{\sin[(2k-1)x]}{(2k-1)^3}. \Rightarrow$$

So we only need to add nonzero (odd n's):

(7) $$x(\pi - x) = \frac{8}{\pi}\sum_{k=1}^{\infty}\frac{\sin[(2k-1)x]}{(2k-1)^3}$$ [Again weird, neither f-n = Σ of odd f-ns!]

(8.16)

(1) $$\frac{2+\pi}{4} + \underbrace{\sum_{n=1}^{\infty}\left[\frac{-1+(-1)^n}{\pi n^2}\cos(nx) - \frac{1-(-1)^n+\pi\cdot(-1)^n}{\pi n}\sin(nx)\right]}_{A}.$$

Let $\boxed{n = 2k}$ $k = 1, 2..$

(2) $$A = \sum_{k=1}^{\infty}\left[\frac{(-1)+(-1)^{2k}}{\pi(2k)^2}\cos(2kx) - \frac{1-(-1)^{2k}+\pi(-1)^{2k}}{\pi(2k)}\sin(2kx)\right] =$$

$$= \sum_{k=1}^{\infty}\left[0 - \frac{\cancel{\pi}}{\cancel{\pi}(2k)}\sin(2kx)\right] = -\sum_{k=1}^{\infty}\frac{\sin(2kx)}{2k}.$$

Let

$\boxed{n = 2k-1}$ $k = 1, 2, \ldots$

$$(3) \quad A = \sum_{k=1}^{\infty} \frac{(-1) + (-1)^{2k-1}}{\pi (2k-1)^2} \cos[(2k-1)x] - \frac{\overset{1-(-1)}{1-(-1)^{2k-1}} + \overset{\pi(-1)}{\pi \cdot (-1)^{2k-1}}}{\pi (2k-1)} \sin[(2k-1)x]$$

$$= \sum_{k=1}^{\infty} \left[\frac{-2}{\pi (2k-1)^2} \cos[(2k-1)x] + \frac{\sin[(2k-1)x]}{(2k-1)} \right] =$$

$$= \sum_{k=1}^{\infty} \frac{-2}{\pi (2k-1)^2} \cos[(2k-1)x] + \sum_{k=1}^{\infty} \frac{\sin[(2k-1)x]}{(2k-1)}$$

We can see in the calculation of (2) & (3) $\sin(nx)$ terms are surviving regardless of choice of n. whereas $\cos(nx)$ terms cancel out when $n = 2k$. So let me separate the given series into 2 & pick the surviving terms as;

$$(4) \quad A = \sum_{k=1}^{\infty} \frac{-2}{\pi (2k-1)^2} \cos[(2k-1)x] + \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(nx)}{n}}_{\text{combining } n = 2k \text{ \& } (2k-1) \text{ cases}}$$

Hence the given series is

$$(5) \quad \frac{2+\pi}{4} + \sum_{k=1}^{\infty} -\frac{2}{\pi (2k-1)^2} \cdot \cos[(2k-1)x] + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot \sin(nx)}{n}.$$

[Compare]

Note that for (8.32) & (8.33), we've had 2 different series.

How can this be? Explain why $\exists$ 2 F.S's for $f(x)$?

- Comparison: $1 \cdot \cos(2kx)$ series in (8.32) has less amplitude than that $\frac{8}{\pi} \cdot \sin[(2k-1)x]$ in (8.33) which is expected, because of the coef-s. So I would say $\sin(\cdot)$ F.S in (8.33) gives a better approximation for longer period.

- How this happens?

$f(x) = x(\pi - x)$ is neither f-n can be written both as a series (sum) of even f-ns & odd f-ns, I guess. But if the given f-n is neither odd nor even f-n, then its F.S must also has similar behaviour.

Below I try to attach the Desmos screen as well.